Coding & Solution

Date	17 July 2026
Project Name	Smart Lender AI: An Intelligent Machine Learning System for Loan Approval Prediction
Maximum Marks	3 Marks

This section evaluates the quality and completeness of the implemented code and the overall solution for Smart Lender AI.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/JinkalaSunitha/Smart-Lender-AI
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, scikit-learn, XGBoost, joblib, Bootstrap 5
Key Features Implemented	ML-based credit risk prediction (XGBoost), client + server-side isolated payload input validation, modern Bootstrap 5 landing interface with async fetch capabilities, and an optimized sub-second real-time scoring engine cache.
Pending / Incomplete Features	Live bank API open-banking ledger ingestion sync, SMS/email alert notification dispatcher routing, and multi-tenant organizational access control role matrices — all tracked as Phase 2 roadmap milestones.
Setup / Run Instructions	pip install -r requirements.txt → python train.py → python app.py → open http://127.0.0.1:5000/. Live cloud deployment setup online framework host gateway.

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

The application has been tested thoroughly end-to-end (home view page interface, prediction data transaction payloads, risk scoring matrices, and wrong parameters/out-of-bounds metrics handling) and executes cleanly across local and staging sandbox pipelines, reading parameters completely in-memory to prevent client logs from leaking.